SUMIT YADAV

Graduate Software Engineer | Python | Django | PostgreSQL

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LinkedIn | GitHub | Portfolio

PROFILE

MSc Advanced Computer Science graduate with over a year of hands-on full-stack project development using Python, Django, JavaScript, PostgreSQL and SQL. Built and deployed CulinaAI with 376 automated tests, CI/CD, AWS S3 and a 10-check validation pipeline. Improved benchmark performance to 30/30 recipes passing first time by fixing validation defects and strengthening food-data matching.

TECHNICAL SKILLS

Programming: Python, JavaScript, PHP, SQL, HTML, CSS
Frameworks and Web: Django, Bootstrap, RESTful API design and integration
Databases: PostgreSQL, MySQL, relational database design, data cleaning and loading
Cloud and DevOps: AWS S3, AWS IAM, Docker, Render, GitHub Actions, CI/CD, Sentry
Testing and Quality: Django TestCase, automated, functional, integration and regression testing
Software Engineering: OOP, System Design, Requirements Engineering, Git, GitHub, GitLab, Agile development
Security: Authentication, RBAC, least-privilege IAM, email confirmation, usage limits, user data controls

SOFTWARE ENGINEERING PROJECTS

CulinaAI | AI-Assisted Recipe Generation and Meal Personalisation System **Jun 2026 - Oct 2026**

MSc Dissertation and Independent Software Engineering Project

Python, Django, PostgreSQL, JavaScript, OpenAI API, AWS S3, Docker, GitHub Actions, Sentry

Live: culinaai.onrender.com | **GitHub:** github.com/SumitYadav2003/culinaai

- Delivered all **29 functional and 10 non-functional requirements** and passed **64/64 structured test scenarios** by translating requirements into Django views, service logic, PostgreSQL models and complete user workflows.
- Built **376 automated Django tests** across recipe generation, validation, accounts, nutrition, meal planning, shopping lists, usage limits and offline behaviour, with GitHub Actions CI/CD running the suite on every pull request and push to main.
- Improved benchmark performance to **30/30 recipes passing quality checks first time** by tracing failed checks, fixing false allergy and equipment matches, improving regeneration feedback and adding regression tests.
- Increased complete ingredient-to-food-data matching from **21/30 to 28/30 benchmark recipes** by expanding ingredient aliases and food mappings for common ingredients and reduced-salt alternatives.
- Built a **10-check validation pipeline** covering allergies, diet, requested ingredients, structure, cooking time, equipment, difficulty, cuisine and health claims, with failed checks passed back for regeneration and a safe fallback when validation still failed.
- Processed **2,759 UK food records** using Python data-cleaning and loading scripts, combining CoFID, ONS, supermarket pricing and carbon data to calculate nutrition, cost and environmental impact in application code.
- Validated **362/362 supported cooking voice-command phrases** by testing navigation, timers, ingredient questions, temperature queries, cooking problems and common wording variations.
- Deployed to production using Render and AWS S3, configuring least-privilege IAM permissions, Sentry monitoring, email confirmation, AI usage limits and account data controls.

CarVerse | Car Marketplace and Dealership Web Application **2023 - 2024**

PHP, MySQL, JavaScript, HTML, CSS

- Built a **40+ page full-stack dealership website** with vehicle listings, multi-image uploads and administrator-controlled CRUD using PHP and MySQL.
- Implemented authentication for **2 user roles** and built database-backed search, comparison, test-drive and service booking, reviews and contact workflows.
- Designed MySQL structures for users, vehicles, bookings, reviews and administrative operations.

EDUCATION

MSc Advanced Computer Science, University of Leicester Sep 2025 - Sep 2026
Dissertation: CulinaAI | AI-Assisted Recipe Generation and Meal Personalisation System

BSc Computer Science, PTVA's Mulund College of Commerce 2021 - 2024
CGPA: **8.53/10**